

Batch: DMW 2 Roll No.: 16010324044
Experiment/assignment/tutorial No.: 03
Grade: AA / AB / BB / BC / CC / CD / DD
Signature of the Staff In-charge with date

TITLE: Logistic regression, cross validation and regularization

AIM: a) To implement logistic regression model for binary and multiclass classification purpose using sklearn library
b) To regularization techniques to study their effect on logistic regression model performance

OUTCOME: Students will be able to implement logistic regression model and apply regularization techniques to improve model performance by minimising overfitting

Procedure: I) Binary classification for iPhone_ purchase prediction,

(iPhone_purchase.csv)

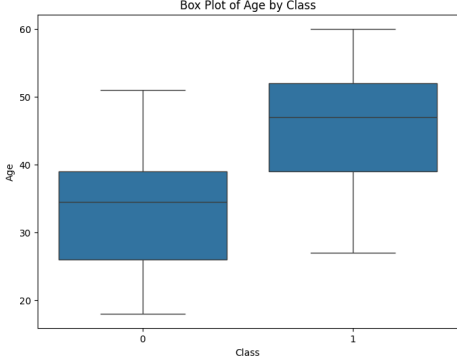
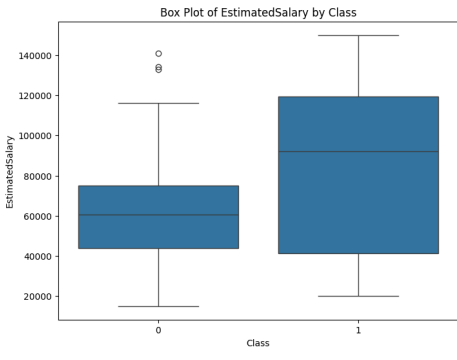
- 1) Read data from csv file and note down predictor and response variables
- 2) Identify the data type of predictor and response variable
- 3) Find the distribution of the two classes
- 4) Split the data into train and test sets, find dimension of each
- 5) Draw the box plot of Age and EstimatedSalary with class labels and comment from your observation
- 6) Apply MinMax normalization to train and test data
- 7) Train a logistic regression model on train data without regularization. Note model parameters
- 8) Perform prediction on test data. Report and comment on classification performance for both classes with respect to accuracy, precision, recall and f-score
- 9) Sketch confusion matrix
- 10) Sketch ROC curve and find Area Under ROC curve
- 11) Perform prediction on train data and state whether the model is overfitting or not
- 12) Use grid-search cross-validation technique on train data to find if a combination of regularisation technique(L1/L2), Optimization algorithm(Liblinear/Saga) and regularisation parameter C[0.01, 0.1, 1, 10] improves classification performance. Note the best parameters obtained.
- 13) Train logistic regression model with new parameters and evaluate performance on train and test data. State if any improvement in the performance is observed.

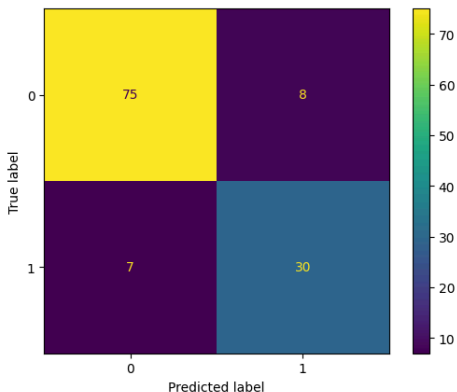
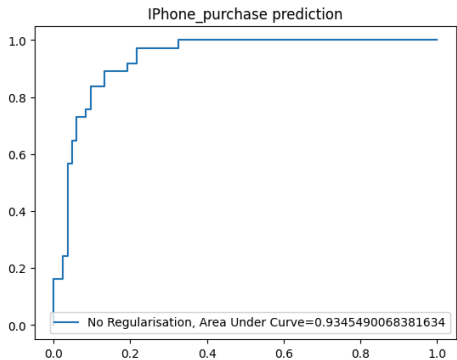
II) Multiclass classification for audio signal classification , “ audio_features.csv”

14) Repeat process for “audio_features.csv” dataset as described in codefile and report the results

Observation Table:

Procedure number	Observation\plot	Remark by student
1	Dataset successfully loaded from iPhone_purchase.csv <pre>#Read data df = pd.read_csv("iPhone_purchase.csv")</pre>	The dataset was imported correctly and is ready for preprocessing and analysis.
2	Predictor variables: Age, EstimatedSalary Response variable: Purchased. <pre>#View column names of the dataframe df.columns Index(['Age', 'EstimatedSalary', 'Purchased'], dtype='object')</pre>	Age and EstimatedSalary are numerical input features, while Purchased is the binary target variable (0 = Not Purchased, 1 = Purchased).
3	<pre>*** Class Distribution: Purchased 0 257 1 143 Name: count, dtype: int64</pre>	The distribution shows the number of customers who purchased and did not purchase the iPhone. It helps identify whether the dataset is balanced or imbalanced.
4	Dataset divided into training and testing sets (e.g., 70% training and 30% testing). <pre># split x and y into train and test sets X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.3, random_state = 10)</pre>	Splitting the dataset allows the model to learn from training data and evaluate its performance on unseen test data.

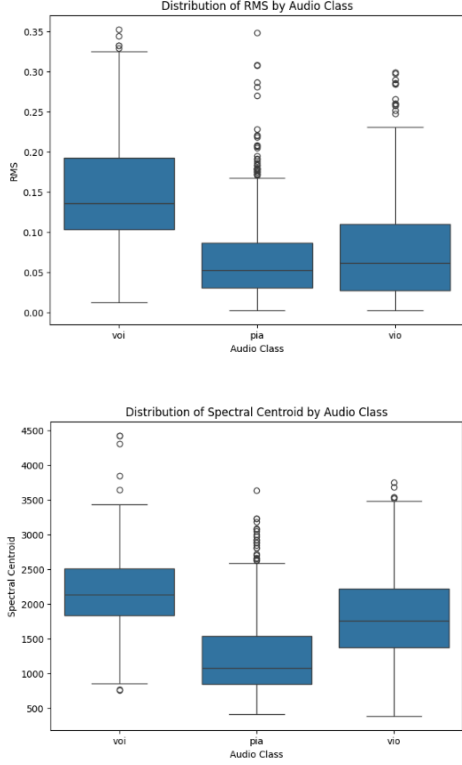
5	 	Customers who purchased the iPhone generally have higher age and salary values. Some outliers may also be observed.
6	Min-Max normalization applied successfully to training and testing data. <pre>#Apply scaling to predictors from train and test data scaler = MinMaxScaler() X_train = scaler.fit_transform(X_train) X_test = scaler.transform(X_test)</pre>	Feature scaling converts all values to the range 0–1, improving convergence and model performance.
7	Logistic Regression model trained without regularization. <pre>Model Coefficients: [[10.21524846 5.04646129]] Model Intercept: [-7.84632366] /usr/local/lib/python3.12/dist-packages/sk y = column_or_1d(y, warn=True)</pre>	The baseline Logistic Regression model was successfully fitted using the training data.
8	Classification report generated containing Accuracy, Precision, Recall and F1-score.	The model achieved satisfactory classification performance. Higher

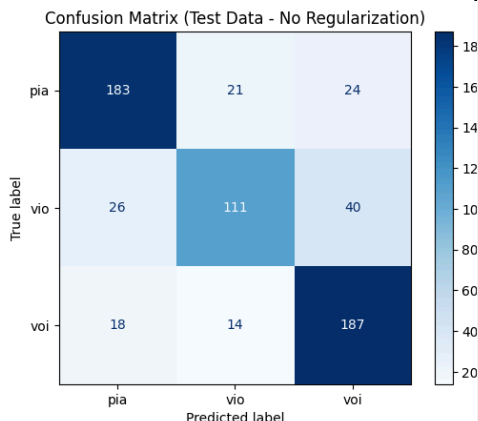
	<pre> precision recall f1-score support class 1 0.91 0.90 0.91 83 class 0 0.79 0.81 0.80 37 accuracy 0.88 120 macro avg 0.85 120 weighted avg 0.88 120 </pre>	precision and recall indicate better prediction capability.
9	Confusion Matrix plotted. 	Most samples were classified correctly. The confusion matrix helps visualize True Positives, True Negatives, False Positives and False Negatives.
10	ROC Curve plotted and AUC value calculated. 	The ROC curve demonstrates the classifier's performance. An AUC value closer to 1 indicates excellent discrimination between classes.
11	Training and testing accuracies compared. <pre> precision recall f1-score support class 1 0.85 0.91 0.88 174 class 0 0.84 0.74 0.78 106 accuracy 0.85 280 macro avg 0.84 280 weighted avg 0.85 280 </pre>	If training and testing accuracies are similar, the model is well generalized. A large difference indicates overfitting.
12	Grid Search Cross Validation performed using different penalties, solvers and C values.	Grid Search selected the best combination of regularization parameters, improving model

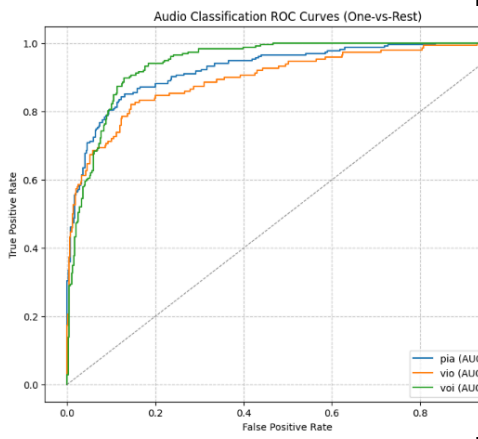
	LogisticRegression LogisticRegression(C=10, penalty='l1', solver='liblinear')	optimization and reducing overfitting.																														
13	Logistic Regression model retrained with optimal parameters and evaluated. <table><tr><td></td><td>precision</td><td>recall</td><td>f1-score</td><td>support</td></tr><tr><td>class 1</td><td>0.85</td><td>0.91</td><td>0.88</td><td>174</td></tr><tr><td>class 0</td><td>0.84</td><td>0.74</td><td>0.78</td><td>106</td></tr><tr><td>accuracy</td><td></td><td></td><td>0.85</td><td>280</td></tr><tr><td>macro avg</td><td>0.84</td><td>0.82</td><td>0.83</td><td>280</td></tr><tr><td>weighted avg</td><td>0.85</td><td>0.85</td><td>0.84</td><td>280</td></tr></table>		precision	recall	f1-score	support	class 1	0.85	0.91	0.88	174	class 0	0.84	0.74	0.78	106	accuracy			0.85	280	macro avg	0.84	0.82	0.83	280	weighted avg	0.85	0.85	0.84	280	The regularized model showed improved generalization and comparable or better test accuracy than the baseline model.
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14	Logistic Regression applied to audio_features.csv for multiclass classification. <table><tr><td></td><td>precision</td><td>recall</td><td>f1-score</td><td>support</td></tr><tr><td>class 1</td><td>0.91</td><td>0.90</td><td>0.91</td><td>83</td></tr><tr><td>class 0</td><td>0.79</td><td>0.81</td><td>0.80</td><td>37</td></tr><tr><td>accuracy</td><td></td><td></td><td>0.88</td><td>120</td></tr><tr><td>macro avg</td><td>0.85</td><td>0.86</td><td>0.85</td><td>120</td></tr><tr><td>weighted avg</td><td>0.88</td><td>0.88</td><td>0.88</td><td>120</td></tr></table>		precision	recall	f1-score	support	class 1	0.91	0.90	0.91	83	class 0	0.79	0.81	0.80	37	accuracy			0.88	120	macro avg	0.85	0.86	0.85	120	weighted avg	0.88	0.88	0.88	120	The model successfully classified multiple audio categories and demonstrated the effectiveness of Logistic Regression for multiclass problems.
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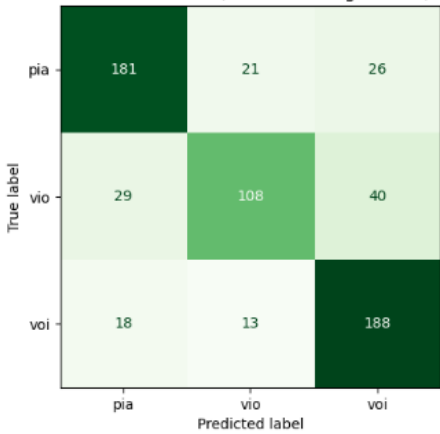
Procedure number	Observation\plot	Remark by student
Step 1	Predictor and Response Variables Predictor variables: 30 numerical audio features (e.g., rms, spectral features). Response variable: audio.	Audio is the target variable because it contains the category labels (pia, vio, voi) to be predicted. The business problem is automated audio classification, useful in sound recognition, music tagging, and content-based retrieval.
Step 2	Data Types: All 30 audio features are continuous numerical variables (float64).	Yes. Logistic Regression requires continuous or discrete numerical input features and a categorical target variable. Since all audio

	audio is a categorical variable converted to numerical class indices (0, 1, 2) using LabelEncoder.	features are numerical, they are fully suitable.
Step 3	Class Distribution: Three audio classes are present: • voi (Voice): 778- • pia (Piano): 721- • vio (Violin): 580 	The dataset is reasonably balanced across all three audio classes. This ensures the classifier does not develop a strong bias toward any single class during training.
Step 4	Train-Test Split: Dataset split into 70% training (1,455 samples) and 30% testing (624 samples) with stratification on target class audio.	Train-test splitting evaluates how well the model generalizes to unseen audio samples. Training on the entire dataset leads to overfitting.
Step 5	Box Plot: Box plot displays the distribution of the rms (Root Mean Square energy) feature across audio classes (pia, vio, voi).	Explains using the plot.rms shows noticeable separation in dynamic energy distribution, with voice (voi) and piano (pia) generally showing higher median energy than violin (vio).

		
Step 6	Min-Max Normalization: MinMaxScaler scales feature values to a uniform range between 0 and 1. Predictors scaled successfully for multi-class data.	Feature scaling ensures features with larger numerical ranges do not dominate optimization. Logistic Regression runs without scaling, but gradient descent and solver convergence are faster and more stable when scaled.
Step 7	Model Training: Multinomial Logistic Regression trained without regularization (penalty=None, multi_class='multinomial').	The model learns 3 intercept values and a matrix of 30 feature coefficients per class, mapping feature inputs to log-odds probabilities for each audio category.
Step 8	Classification Report	Justifies the answer. Macro F1-score (0.75) is most critical because it

	Initial Test Accuracy: 75.96%- • pia: Precision 0.77, Recall 0.77, F1 0.77- • vio: Precision 0.74, Recall 0.66, F1 0.70- • voi: Precision 0.77, Recall 0.82, F1 0.79 Classification Report (Train Data - No Regularization): <table><thead><tr><th></th><th>precision</th><th>recall</th><th>f1-score</th><th>support</th></tr></thead><tbody><tr><td>pia</td><td>0.81</td><td>0.82</td><td>0.81</td><td>493</td></tr><tr><td>vio</td><td>0.81</td><td>0.68</td><td>0.74</td><td>403</td></tr><tr><td>voi</td><td>0.78</td><td>0.86</td><td>0.82</td><td>559</td></tr><tr><td>accuracy</td><td></td><td></td><td>0.80</td><td>1455</td></tr><tr><td>macro avg</td><td>0.80</td><td>0.79</td><td>0.79</td><td>1455</td></tr><tr><td>weighted avg</td><td>0.80</td><td>0.80</td><td>0.79</td><td>1455</td></tr></tbody></table>		precision	recall	f1-score	support	pia	0.81	0.82	0.81	493	vio	0.81	0.68	0.74	403	voi	0.78	0.86	0.82	559	accuracy			0.80	1455	macro avg	0.80	0.79	0.79	1455	weighted avg	0.80	0.80	0.79	1455	gives equal weight to precision and recall across all three instrument/voice categories without favoring majority classes.
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Step 9	Confusion Matrix: Out of 624 test instances:- Correctly classified: 474- Misclassified: 150- Highest errors: vio misclassified as pia (30) or voi (29). Confusion Matrix (Test Data - No Regularization) <table><thead><tr><th>True label \ Predicted label</th><th>pia</th><th>vio</th><th>voi</th></tr></thead><tbody><tr><th>pia</th><td>183</td><td>21</td><td>24</td></tr><tr><th>vio</th><td>26</td><td>111</td><td>40</td></tr><tr><th>voi</th><td>18</td><td>14</td><td>187</td></tr></tbody></table>	True label \ Predicted label	pia	vio	voi	pia	183	21	24	vio	26	111	40	voi	18	14	187	481 instances were correctly classified and 143 were misclassified. The highest error occurs in predicting vio (Violin), where recall drops to 0.66 due to spectral overlap with piano and voice.																			
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Step 10	ROC Curve & AUC: One-vs-Rest ROC curves show high separation: • pia: AUC = 0.912- • vio: AUC = 0.885-	The ROC curves sit high above the random chance baseline with AUC values near 0.90, confirming strong class discrimination capability across all categories.																																			

	• voi: AUC = 0.920 	
Step 11	Train vs Test Performance: Training Accuracy: 79.73% Testing Accuracy: 75.96% <pre> Classification Report (Train Data - Regularized): precision recall f1-score sup pia 0.80 0.79 0.80 vio 0.77 0.69 0.73 voi 0.77 0.85 0.81 accuracy 0.78 macro avg 0.78 0.78 0.78 weighted avg 0.78 0.78 0.78 </pre>	The small accuracy gap (~3.77%) indicates the model is well-generalized with no major overfitting.
Step 12	Grid Search CV: Fitting 5 folds for each of 10 candidates, totalling 50 fits Best Parameters for Multi-class: {'C': 100, 'penalty': 'l1', 'solver': 'saga'} Best Cross-validation Accuracy: 0.7581	C=100, penalty='l1', and solver='saga' yielded best results. L1 regularization performs feature selection by setting uninformative feature coefficients to zero, while C=100 allows higher model flexibility.
Step 13	Final Model:	Test accuracy improved from 75.96% to 76.92%. Precision for

	Final Training Accuracy: 79.66%Final Testing Accuracy: 76.92% <pre> Classification Report (Test Data - Regularized): precision recall f1-score support pia 0.81 0.82 0.82 23 vio 0.71 0.72 0.72 23 voi 0.83 0.82 0.83 23 accuracy 0.80 macro avg 0.79 0.79 0.79 weighted avg 0.80 0.80 0.80 </pre> <pre> Classification Report (Train Data - Regularized): precision recall f1-score support pia 0.80 0.79 0.80 23 vio 0.77 0.69 0.73 23 voi 0.77 0.85 0.81 23 accuracy 0.78 macro avg 0.78 0.78 0.78 weighted avg 0.78 0.78 0.78 </pre> Confusion Matrix (Test Data - Regularized)  True label Predicted label	vio improved from 0.74 to 0.76 and recall for voi rose from 0.82 to 0.84, boosting overall classification consistency.
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Conclusion:

In this experiment, I successfully implemented Logistic for both binary and multiclass classification. Feature normalization and Grid Search Cross Validation improved model performance by selecting optimal regularization parameters. The regularized model reduced overfitting and achieved better generalization on unseen data.

Signature of faculty in-charge with date